



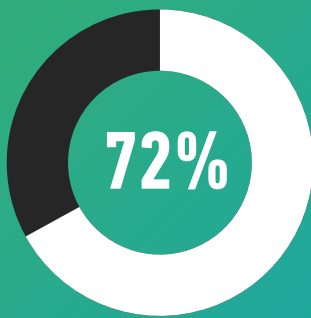
WHITEPAPER

REVOLUTIONIZING RETAIL WITH IMAGE RECOGNITION AND AI FOR SHELF EXECUTION



Artificial intelligence (AI) is changing the way the world operates. Businesses are harnessing innovation and cutting-edge technologies to optimize operations, enhance decision-making processes, and fuel business growth.

Recent research indicates a significant growth in technology adoption rate within the retail industry, particularly in AI and image recognition technology. By the end of 2035, global image recognition in the retail industry is projected to reach a revenue of \$14 billion.¹ This growth is attributed to factors such as the demand for automated processes, the utilization of AI and cloud technologies, and the need for personalized customer experiences. In North America, the market is expected to generate the largest revenue due to technological advancements and the implementation of image recognition solutions.



72% of retailers using AI experienced a decrease in operating costs

Image recognition in the retail market is projected to grow at a 23% CAGR from 2023 to 2033

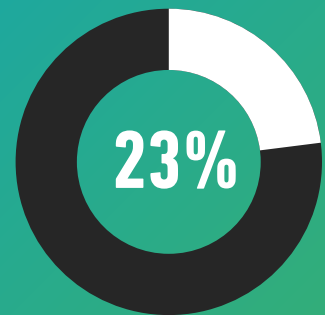


Image 1

These powerhouse innovations are coming together in PhotoAI to revolutionize how retailers approach shelf execution, predict consumer behavior, and enhance operational efficiency. This game-changing duo is reshaping the landscape of in-store operations, allowing store and field teams to operate strategically and effectively.

¹Global News Wire, <https://www.globenewswire.com/en/news-release/2023/07/25/2710327/0/en/Image-Recognition-in-Retail-Market-Revenue-to-Cross-USD-14-Billion-by-2035-says-Research-Nester.html>. Accessed 10 March 2024.

CHALLENGES RETAILERS ARE FACING TODAY

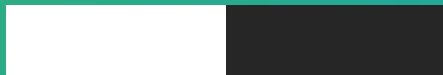
The industry has a lengthy history of operational challenges, from manual inventory checks to labor-intensive merchandising processes. Conventional methods are not only time-consuming but also prone to errors, hindering the overall efficiency of businesses. Constantly grappling with inaccurate inventory data, misplaced products, and inefficient shelf execution wastes valuable time and resources and results in missed sales opportunities and dissatisfied customers.



Phantom inventory cost
retailers 4% in annual revenue



Stockouts have increased to
more than 35%



Manual audit processes have
a 50% error rate



91% of customers won't engage
with a store after a bad experience

Image 2

The increased pace of technological innovation means that retailers must constantly adapt and evolve their operations to keep up with changing trends and consumer behaviors. PhotoAI addresses these challenges with proactive inventory monitoring and accurate demand forecasting, revolutionizing shelf execution and replacing traditional manual processes with real-time, unbiased solutions that deliver actionable insights.

THE POWER OF SHELF EXECUTION

Shelf execution is not just a necessary component of retail operations; it drives customer experience and fuels retail profitability. Strategic product placement, meticulous adherence to planograms, and ensuring optimal on-shelf availability can create an irresistible shopping environment that sparks impulse purchases and drives revenue.

Encountering out-of-stock items, misplaced products, or pricing discrepancies due to poor shelf execution or item management undermines store reputation and brand

perception in customers' eyes. These slip-ups erode trust, diminish satisfaction, and result in lost sales opportunities. Whereas well-executed shelf management can reflect a retailer's commitment to excellence and attention to detail. Seamless and successful shopping experiences leave positive impressions, fostering customer satisfaction, loyalty, and repeat business. Shelf execution is a tangible manifestation of brand identity and values, signaling professionalism, reliability, and trustworthiness to customers.

Exceptional shelf management practices are more than a routine task; they're an irrefutable strategic imperative for success. By prioritizing accuracy, consistency, and efficiency in product placement, both store and field teams can enhance profitability and foster long-lasting relationships with customers. In the age of omnichannel retailing and ever-evolving consumer preferences, mastering the art of shelf execution is essential for retailers looking to thrive and succeed in today's dynamic marketplace.

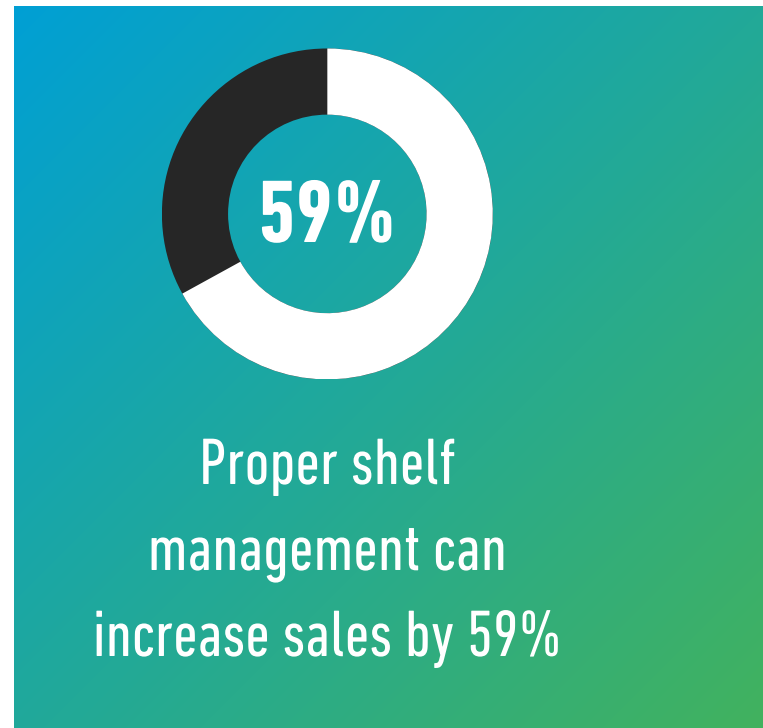


Image 3

COMPONENTS OF SHELF EXECUTION



On-Shelf Availability

From Out-of-Stock Alerts to "In-Stock but Not on Shelf." Direct ties with automated tasking and integrated across all retail stakeholders.



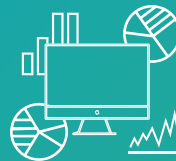
Photo Quality Checks

Enforce adherence to minimally acceptable standards for photos in terms of quality and orientation.



Negative Object Detection

Detect if an image contains an unwanted object; human hand, a human, the floor, etc.



Share of Shelf

Calculation of your category ownership compared to your competitors.



Planogram Compliance

Scored feedback (in-aisle or batch-processed) on how much an image matches the base image / ideal planogram.



Pricing Compliance

Automatically detect if the price listed on a shelf label matches what the retailer has in their database.



Retail Banner Distribution

Calculation of product distribution percentage across all stores in a retail banner, or across banners.



AI Tasking

Tasks automatically generated from system-identified issues in analyzed photos (e.g. re-stock required).



Trade Fund Compliance

Retailers need to protect the money brands spend with them to get incremental placements and other concessions. Automated photo verification of promised concessions keep trade funds safely with the retailer.

RISE OF IMAGE RECOGNITION TECHNOLOGY INTO RETAIL

Image recognition has humble beginnings that can be traced back to the early days of artificial intelligence research. In the 1950s and 1960s, scientists began exploring ways to teach computers to "see" and interpret visual data. Early experiments laid the groundwork for modern image recognition algorithms, which have since become a cornerstone of many industries, including retail.

In recent years, image recognition technology has undergone significant advancements, leading to pivotal moments in its development and widespread adoption across various industries, including retail.



Walmart

Walmart has been actively investing in image recognition technology to optimize operations, monitor shelf conditions in real-time, and inventory management to track product availability and replenishment needs.

Integrating artificial intelligence with image recognition represents a paradigm shift in retail shelf execution. Leveraging machine learning to analyze visual data in real-time revolutionizes how stores manage their shelves and operations.

Image recognition technology allows computers to interpret and understand visual information from images and videos. By training algorithms on vast datasets of product images, shelf layouts, and planograms, image recognition systems can accurately identify products, assess shelf conditions, and detect anomalies with precision and speed. This capability enables retailers to automate critical tasks such as shelf audits, inventory monitoring, and planogram compliance checks, reducing manual labor and human error.



Procter & Gamble

P&G uses image recognition technology to monitor product placements and compliance with planograms in retail stores to ensure its products are prominently displayed and easily accessible to customers, maximizing sales opportunities.

Image recognition technology integrated with artificial intelligence learns from experience and adapts to new inputs to perform tasks that typically require human intelligence. AI algorithms can analyze extensive amounts of data, recognize complex patterns, and make informed decisions in real-time, enabling retailers to optimize shelf execution processes and drive operational efficiency.

Instead of relying solely on manual inspections or periodic audits, store and field teams can monitor shelf conditions continuously and in real-time, ensuring that products are always stocked, organized, and priced correctly.



Coca-Cola

Coca-Cola analyzes images of displays to ensure its products are prominently featured and properly stocked, driving sales and brand visibility.

Additionally, these technologies facilitate data-driven decision-making, allowing retail teams to optimize product placement, adjust inventory levels, and tailor shelf layouts to meet changing consumer preferences and market trends. Managers can identify patterns, understand shopper behavior, and make strategic adjustments to drive sales and profitability by analyzing customer interactions with products and shelves.



Nestlé

Nestlé utilizes image recognition technology to identify opportunities for optimizing product placement, tailoring promotional campaigns, and enhancing the overall shopping experience for customers.

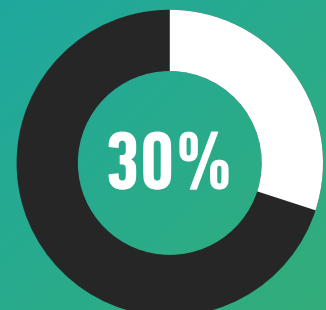
By automating critical tasks, providing valuable insights, and enabling data-driven decision-making, these technologies empower retailers to enhance shelf execution processes, improve the customer experience, and drive business growth in today's dynamic retail landscape.

PHOTOAI IN RETAIL

At its core, PhotoAI leverages the combined power of image recognition and AI algorithms to analyze visual data from store shelves in real-time. By automatically detecting products, assessing shelf conditions, and ensuring compliance with planograms, PhotoAI enables stores to maintain optimal on-shelf availability, improve product placement, and enhance the overall customer experience.

PhotoAI is revolutionizing shelf and merchandise management across grocery chains, department stores, convenience stores, and specialty retailers. By serving as a gateway to valuable insights into shelf conditions, product performance, and customer behavior, PhotoAI is an invaluable ally, empowering retail teams of all kinds to optimize product assortments, refine marketing strategies, and ignite unprecedented sales growth.

Photo AI enables companies to analyze visual data effortlessly, saving up to 30% of time and resources.



DIGITAL TWINS IN SHELF MANAGEMENT

Agile companies are more immune to market volatility. Embracing technology to equip inventory planners with increased agility helps teams swiftly adapt to changing demands and safeguard margins in any situation. By harnessing the power of PhotoAI within their digital twin ecosystems, retailers can future-proof their businesses and navigate the complex challenges of the retail landscape.

Digital twins enable retailers to optimize operations and customer experiences by creating virtual replicas of physical objects, assets, or environments. They have become a cornerstone of strategic planning for retailers, empowering them with real-time and predictive data crucial for decision-making.

There is a deluge of data available for retailers to tap into, but without the appropriate tools, it can be challenging to transform that data into actionable strategies. Digital twin technology augmented by PhotoAI offers a virtual model for testing various scenarios, particularly in strategic inventory and merchandise planning.

Creating virtual replicas of physical operations gives retail teams a comprehensive view of their shelf management practices. Armed with these insights, inventory and demand planners can craft adaptable, data-informed strategies to model various what-if scenarios and confidently tackle unexpected changes or challenges that will inevitably continue to shape their industry.



**DIGITAL TWIN
TECHNOLOGY
IS EXPECTED
TO GROW \$53
BILLION BY
2028**

ENHANCED SHELF EXECUTION FOR BETTER IN-STORE EXPERIENCE

Shelf execution is more than just filling up the shelves and attending to customers—it's about meticulousness, effectiveness, and forward-thinking. Bringing PhotoAI into the mix opens up a world of advantages that streamline tasks, boost efficiency for store and field teams, and increase profits. With its real-time shelf analysis, precise demand forecasting, and quick identification and resolution of retail hurdles, PhotoAI equips teams with the tools they need to fine-tune their operations and offer top-notch customer service.

PhotoAI can spot

- **Out-of-stock products**
- **Product misplacements**
- **Pricing inconsistencies**

PhotoAI promptly highlights disparities by cross-referencing actual shelf conditions with the anticipated state to empower store and field teams to tackle issues swiftly. Powered by machine learning algorithms that continually evolve through experience, it enhances accuracy in detecting issues while minimizing erroneous alerts. AI enables retailers to restock shelves, reposition items, or rectify pricing discrepancies. These actions improve operational efficiency and enhance the overall customer experience by ensuring that shelves are always stocked with the right products in the right place at the right time.





DATA-DRIVEN MERCHANDISING INSIGHTS

PhotoAI revolutionizes the retail landscape by providing retailers with invaluable merchandising insights derived from diverse data sources, ultimately empowering strategic decision-making and enhancing the customer shopping experience. By harnessing the power of advanced data analytics, PhotoAI collects and dissects a wealth of data from various sources, such as purchasing trends, correlations, and patterns, to guide strategic merchandising choices, product selection strategies, and promotional campaigns. With its cutting-edge analytical capacities, retailers gain deep insights into customer preferences, enhance product placements for optimal results, and craft personalized promotions to boost sales revenue. Leveraging PhotoAI's insights leads to informed decision-making grounded in data analysis that elevates the impact of merchandising endeavors while enriching the shopping journey.

REAL-TIME SHELF ANALYSIS

Harnessing the power of advanced computer vision technology, PhotoAI revolutionizes retail operations by providing real-time insights into store shelf images. It accurately identifies product placements, inventory levels, and planogram adherence by detecting products and their locations on shelves. Machine learning algorithms distinguish between various products, packaging types, and typical shelf arrangements. This analysis generates actionable insights for retailers to enhance shelf layouts, streamline inventory control, and maintain merchandising standards. Retailers can take immediate corrective actions by identifying these issues in real-time, minimizing lost sales opportunities, and maximizing revenue.

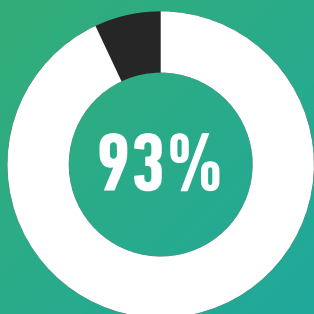


INSTANT COMPLIANCE VISIBILITY

Integrating PhotoAI into shelf management delivers instant compliance visibility by swiftly scanning shelf images to pinpoint and highlight potential compliance concerns such as empty shelves, misplaced items, and deviations from planograms or promotional displays. With real-time insights into these critical areas, store and field teams can expediently identify and address any compliance issues that arise to ensure that merchandising standards are consistently upheld. This proactive approach not only streamlines decision-making processes but also enables teams to optimize shelf layouts effectively, significantly boosting overall performance and customer satisfaction.

ACCURATE DEMAND FORECASTING AND SHELF STOCKING

By integrating AI with image analysis, PhotoAI revolutionizes retail operations by predicting demand, optimizing inventory levels, and delivering personalized shopping experiences based on real-time data insights. PhotoAI utilizes historical sales data and shelf analysis to predict demand with precision. Analyzing customer interactions with products and shelves helps to identify trends, preferences, and opportunities for optimization, enabling retail teams to deliver more personalized and relevant shopping experiences. Through image analysis of products and displays, it evaluates existing inventory levels and detects shortages or excess stock. Machine learning algorithms continuously enhance forecasting accuracy by integrating real-time data and learnings from past forecasts. Retail teams benefit from tailored stocking recommendations for optimal inventory levels and strategic placement, resulting in streamlined processes that minimize stockouts and overstock scenarios.



Machine learning can provide
seasonal demand forecasting with an
accuracy rate of 93%

MINIMIZING ERRORS AND STREAMLINING WORK PROCESSES

Audit and inspection tasks have traditionally been time-consuming and labor-intensive for store teams. PhotoAI effectively reduces manual entry errors by accurately identifying product placements and ensuring compliance with planogram standards. This automation can significantly reduce the need for manual inspections, decreasing errors and manual tasks while optimizing operations, improving efficiency, and increasing productivity levels. This boost in operational efficiency enhances resource management, ultimately leading to superior performance and resource utilization.

Retail has a 60% labor turnover rate,
the highest of any industry

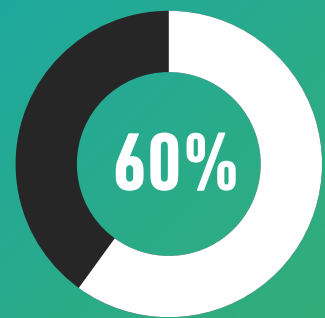


Image 7

MANAGING WORKLOADS

The need to manage workloads in the retail sector has become increasingly pronounced in today's high-turnover work environment. With ever-growing demands on retail teams to deliver exceptional customer experiences while simultaneously juggling multiple operational tasks, the pressure to streamline processes and enhance efficiency has never been greater. The imperative for efficiency is further compounded by fluctuating consumer preferences, evolving market trends, and the relentless pace of technological innovation.

THE FUTURE OF RETAIL

As technology advances, the shopping experience is evolving with endless opportunities ahead. Integrating artificial intelligence and image recognition technology is poised to revolutionize how retail teams operate and gain valuable insights, paving the way for a new era in retail operations.

SHELF EXECUTION IS A CORNERSTONE OF RETAIL SUCCESS, PLAYING A CRITICAL ROLE IN SHAPING CUSTOMER EXPERIENCES, DRIVING SALES, AND MAINTAINING BRAND INTEGRITY.

How products are presented and organized on shelves directly impacts customer perception, purchase decisions, and, ultimately, the bottom line. Effective shelf execution ensures that products are readily accessible, visually appealing, and accurately priced, fostering positive customer interactions and maximizing sales opportunities.

However, achieving optimal shelf execution poses numerous challenges for retail teams. Ensuring that shelves adhere to planograms can be a daunting task. Managing inventory levels and ensuring stock availability on shelves can also be challenging, requiring management to invest heavily in resources to find the balance between overstocking and stockouts.

Determining and regulating the optimal placement of products on shelves to maximize visibility is a constant struggle for retail teams. Product size, packaging, and shelf positioning all contribute to influencing customer interest and buying habits. It's crucial to accurately display product prices on shelves to uphold customer trust and prevent discrepancies that may otherwise lead to customer dissatisfaction and lost sales opportunities.

FACING RETAIL CHALLENGES HEAD ON

Advancements in technology are actively addressing the myriad of challenges surrounding shelf execution and inventory management. The complexities are manifold, from maintaining planogram compliance to optimizing inventory levels. The combination of PhotoAI and digital twin technology presents retailers with a valuable edge in the realm of shelf management and beyond. By fully embracing these advanced solutions to harness the complete power of their data, retail teams can respond effectively to shifting market trends, boost their operational efficiency, and set the stage for a more organized and customer-centric shopping journey.

Equipped with advanced machine learning algorithms, PhotoAI swiftly identifies out-of-stock products, misplaced items, and pricing discrepancies in real-time. Providing instant alerts and actionable insights empowers personnel to take swift measures to rectify issues, ensuring that shelves are always stocked with the right products in the right place at the right time.

PhotoAI's ability to conduct instant audits, monitor compliance in real-time, and provide actionable insights derived from advanced analytics revolutionizes how store and field teams operate. Management can empower their teams with real-time, unbiased solutions for enhancing shelf execution, ultimately improving operational efficiency and customer satisfaction.



Image Citation

Image 1

Countouris, Cynthia. "Survey Reveals Latest Trends Driving Technological Advancements in Retail Industry." NVIDIA Blog, 9 January 2024, <https://blogs.nvidia.com/blog/ai-in-retail-survey-2024/>. Accessed 10 March 2024.

Saha, Sudip. "Image Recognition in Retail Market Share, Size & Trends 2033." Future Market Insights, <https://www.futuremarketinsights.com/reports/image-recognition-in-retail-market>. Accessed 10 March 2024.

Image 2

Farias, Vivek F., et al. "Fixing Inventory Inaccuracies At Scale." ArXiv, <https://arxiv.org/pdf/2006.13126.pdf>. Accessed 10 March 2024.

Knight, Benjamin, and Dmitry Mitrofanov. "Why You Should Warn Customers When You're Running Low on Stock." Harvard Business Review, 5 September 2022, <https://hbr.org/2022/09/why-you-should-warn-customers-when-youre-running-low-on-stock>. Accessed 10 March 2024.

Zobrist, Teresa. "How Running Out-of-Stock Affects Your Consumers." Smart Merchandiser, 31 May 2021, <https://www.smartmerchandiser.com/out-of-stock-affects-customers/>. Accessed 10 March 2024.

Image 3

Pontigo Marín, Sofia V. "Measuring the impact of shelf position on a retailer's sales." Repositorio UCHILE, 2021, <https://repositorio.uchile.cl/bitstream/handle/2250/180420/Measuring-the-impact-of-shelf-position-on-a-retailer-s-sales.pdf?sequence=1&isAllowed=y>. Accessed 10 March 2024.

Image 4

Nielsen, Jakob. "AI Improves Employee Productivity by 66%." Nielsen Norman Group, 16 July 2023, <https://www.nngroup.com/articles/ai-tools-productivity-gains/>. Accessed 10 March 2024.

Image 5

Vantage Market Research. Global News Wire, 2022, <https://www.globenewswire.com/news-release/2022/08/26/2505314/0/en/Global-Digital-Twin-Market-Size-Share-to-Surpass-USD-53-5-Billion-by-2028-at-a-CAGR-42-1-Vantage-Market-Research.html>. Accessed 10 March 2024.

Image 6

Springer US. "A data-driven forecasting approach for newly launched seasonal products by leveraging machine-learning approaches." Massachusetts Institute of Technology, 2020, https://dspace.mit.edu/bitstream/handle/1721.1/136919/10479_2020_3666_ReferencePDF.pdf?sequence=1&isAllowed=y. Accessed 10 March 2024.

Image 7

Business of Labor Statistics. "Job Openings and Labor Turnover - January 2024." Bureau of Labor Statistics, 6 March 2024, <https://www.bls.gov/news.release/pdf/jolts.pdf>. Accessed 10 March 2024.